长视频模型 / 世界模型的
KV Cache量化

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Outline

01 生成范式与KV Cache瓶颈

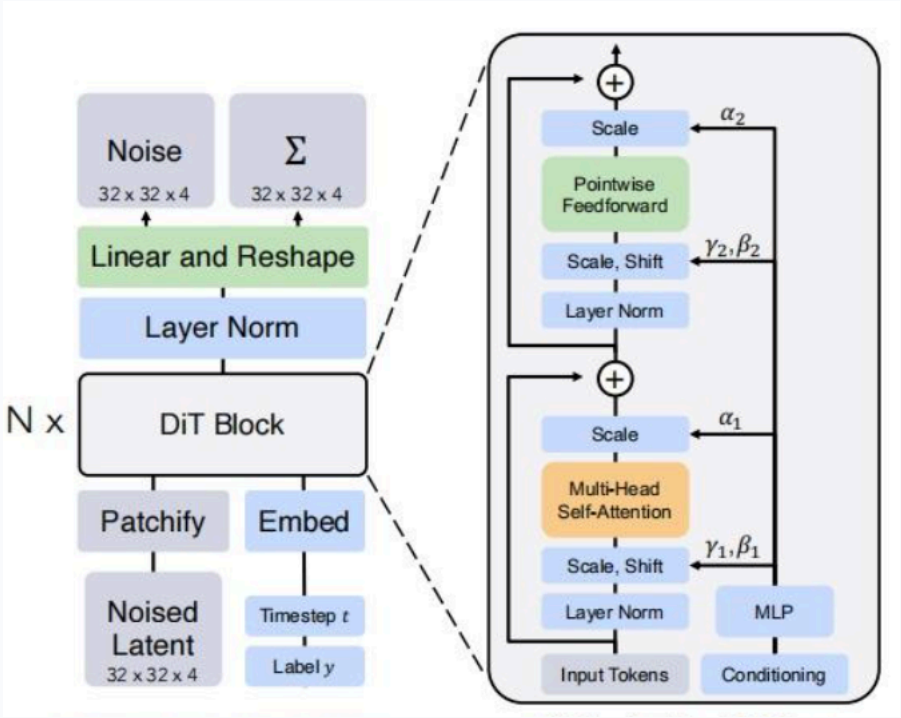
02 QVG、Bias Correction、QuantWM

[ICML 2026]Quant VideoGen: Auto-Regressive Long Video Generation via 2-Bit KV-Cache Quantization

[ICML 2026 SCALE & F2S Workshops]Quantized Keys Steal Attention: Bias Correction for KV-Cache Compression in Video Diffusion

QUANTWM: TEMPORALLY CONSISTENT 2-BIT KV CACHE QUANTIZATION FOR WORLD MODELS AND VIDEO GENERATION

生成范式变化



Diffusion Transformer

Video

x N timesteps

Dit进行N个timestep去噪生成整段视频

Chunk 1
去噪完成

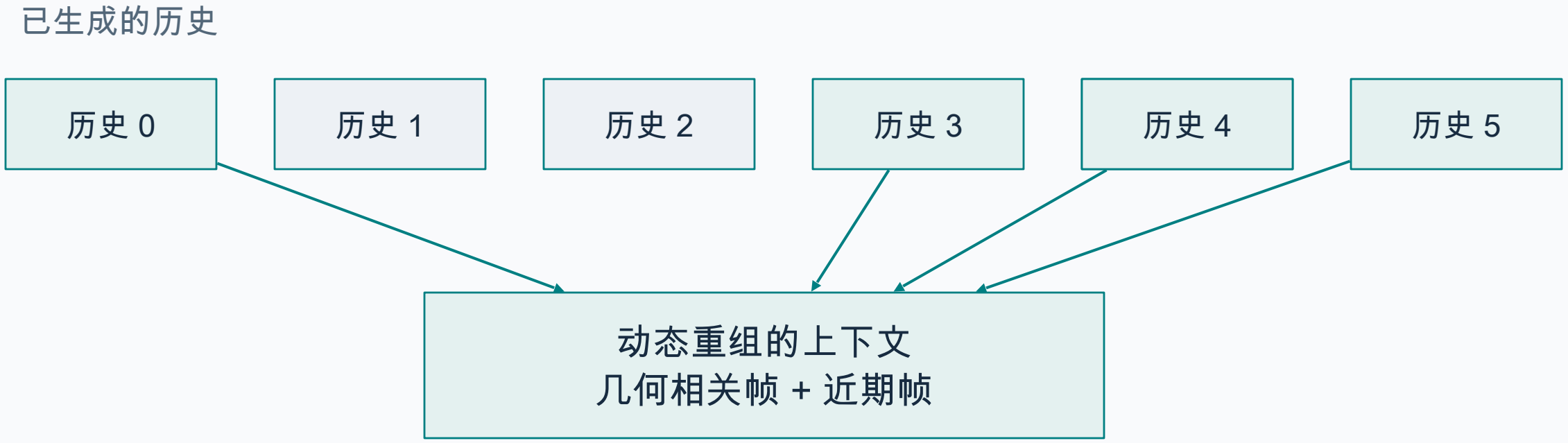
Chunk 2
正在去噪

Chunk 3
待生成

x N timesteps　x N timesteps　x N timesteps

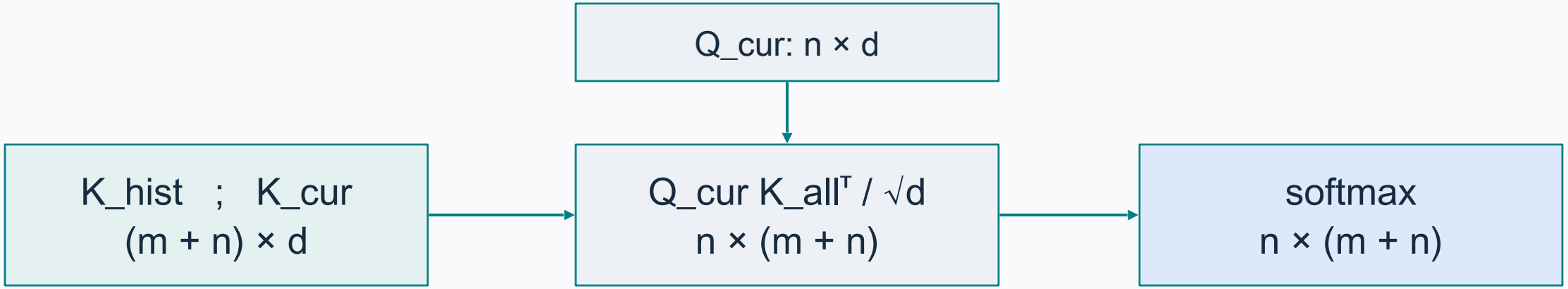
一段视频分成多个chunk，逐chunk生成，
先前chunk通过KV Cache的方式参与当前
chunk的生成

KV Cache组成（以HY-World1.5为例）



视频生成前会先生成相机路径，通过路径得到的几何相关帧与近期帧（有的模型会保留第一帧）得到当前chunk的KV Cache。
在Diffusion loop外完成KV Cache的选取，一个chunk生成中KV Cache保持不变

KV拼接与当前Chunk Attention



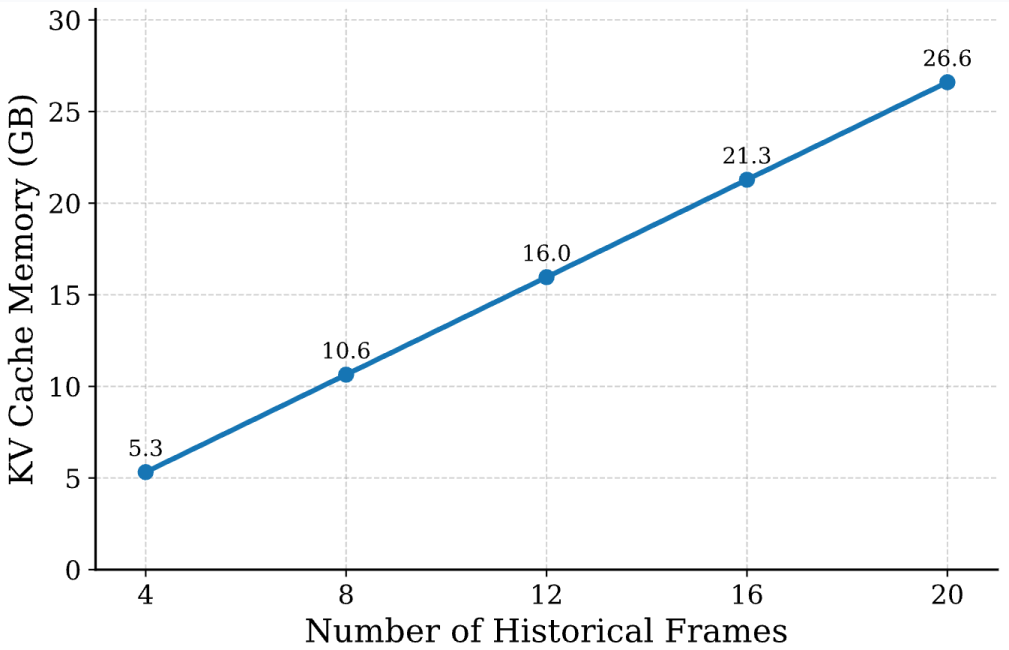
$$K_{all} = [K_{hist}; K_{cur}] \ , \ V_{all} = [V_{hist}; V_{cur}]$$
$$O = softmax(Q_{cur}K_{all}^{T}/\sqrt{d} + mask)V_{all}$$

沿token维拼接KV

为什么需要量化KV Cache

存储开销： 长视频历史 KV Cache 占用大量显存；

访存开销： 每次生成新 chunk 时需要从 KV Cache 读取大量历史 K/V；



Mcache = 2·L·N·H_kv·d_head × b / 8 + Mmetadata

HY-World 1.5(5 chunk 20 frames)：2 x 56层 × 31200 tokens × 16 x 128

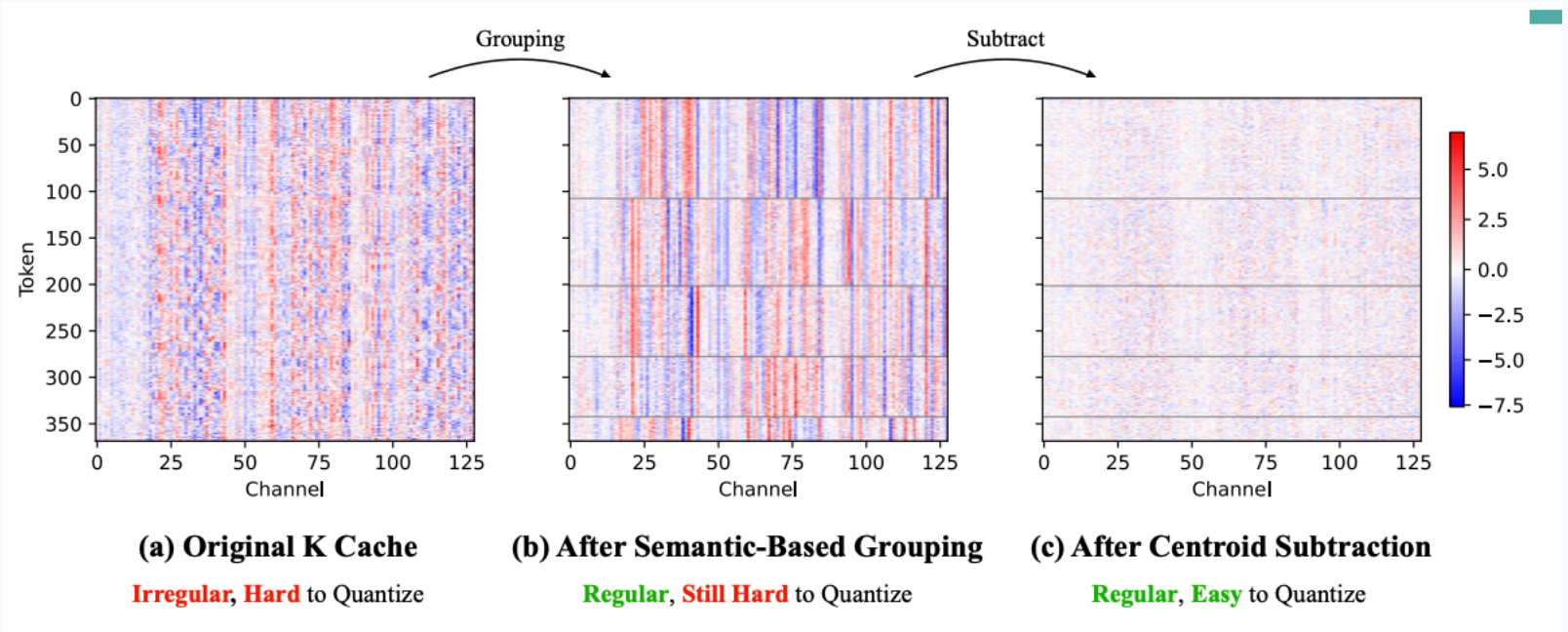
BF16 26.6 GiB

Quant VideoGen：视频冗余与残差压缩

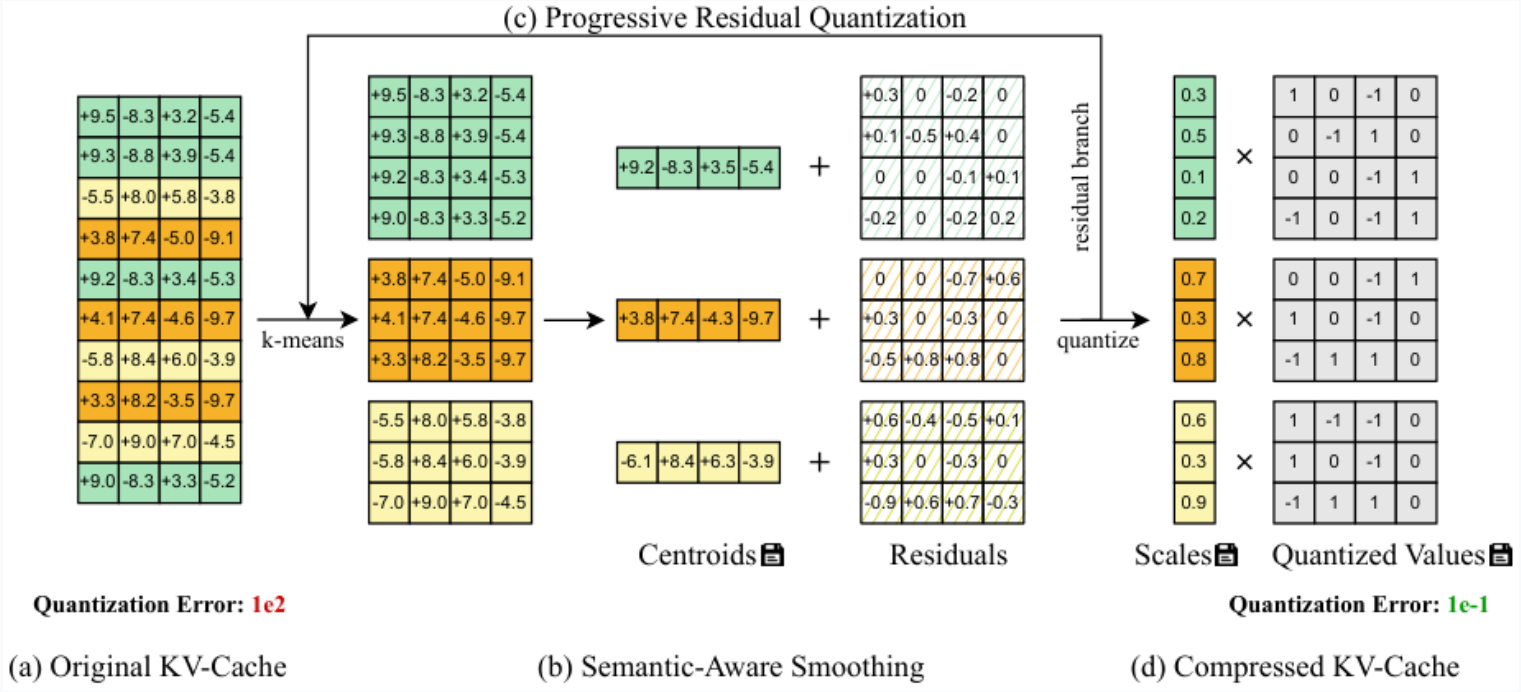
training-free，逐chunk压缩历史KV

Semantic-Aware Smoothing降低残差动态范围

Progressive Residual Quantization逐级细化表示

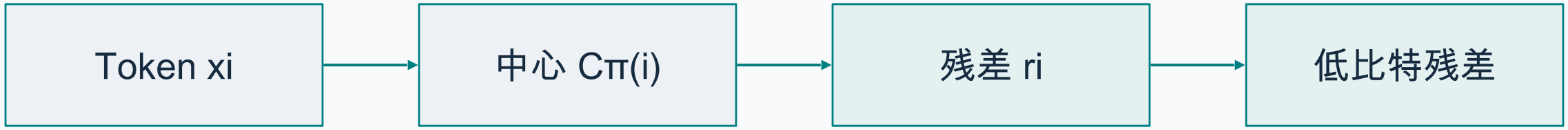


Semantic-Aware Smoothing（SAS）



顺序：Kmeans聚类相似token、每类减去聚类中心、量化残差
centroid与assignment也是缓存的一部分，但是由于一类只需要存一个centroid，每个token存一个index，开销平摊下来不是特别大

Semantic-Aware Smoothing



$$r_i = x_i - C_\pi(i)$$
$$\hat{x}_i = C_\pi(i) + Dequant(Quant(r_i))$$

沿token维对每个chunk的KV表示做k-means

残差幅值变小，低比特网格更有效

成本：中心、索引与在线聚类

Progressive Residual Quantization

每一级对上一级残差重新聚类并减中心

只量化最终残差，保存各级中心与索引

增加阶段改善质量，同时增加元数据

$$R^{(0)} = X \qquad R^{(t)} = R^{(t-1)} - C^{(t)}[\pi^{(t)}]$$

$$\hat{X} = \mathrm{Dequant}(\mathrm{Quant}(R^{(T)})) + \Sigma t\ C^{(t)}[\pi^{(t)}]$$

QVG : Result

Table 1. Quality and Compression results of QVG and baselines. QVG-Pro achieves $4.97\times \sim 5.20\times$ compression ratio, while achieving much better fidelity scores than all baselines. QVG further pushes the compression ratio to $6.94\times \sim 7.05\times$ and still maintains near-lossless video quality scores.

Method	Compression Ratio (BF16)	PSNR	SSIM	LPIPS	Background Consistency	Image Quality	Subject Consistency	Aesthetic Quality
LongCat-Video-13B		*INT2 KV Cache, 480p*			96.22	72.72	95.51	64.83
RTN	6.40×	20.872	0.719	0.203	84.84	59.60	70.63	43.38
KIVI	6.40×	20.317	0.719	0.208	84.84	38.10	75.25	41.58
Quarot	6.40×	21.573	0.759	0.171	86.12	50.70	80.61	49.49
QVG-Pro	4.97×	**30.376**	**0.935**	**0.048**	96.20	71.74	94.92	63.88
QVG	**6.94**×	28.716	0.909	0.065	95.06	71.47	94.11	62.22
LongCat-Video-13B		*INT4 KV Cache, 480p*			96.22	72.72	95.51	64.83
RTN	3.55×	32.984	0.940	0.045	96.13	72.14	95.53	64.57
KIVI	3.55×	32.158	0.946	0.040	96.16	72.67	95.26	64.53
Quarot	3.55×	33.744	0.960	0.033	95.48	72.47	95.48	64.92
QVG-Pro	3.05×	**37.095**	**0.977**	**0.024**	96.67	72.66	95.44	64.93
QVG	**3.72**×	37.141	0.978	0.024	95.94	72.34	94.34	64.88
HY-WorldPlay-8B		*INT2 KV Cache, 480p*			97.92	74.33	97.90	69.85
RTN	6.40×	24.199	0.696	0.229	96.16	71.86	96.08	69.15
KIVI	6.40×	24.272	0.701	0.230	96.95	71.40	95.89	68.19
Quarot	6.40×	25.207	0.738	0.205	97.34	72.26	96.64	69.38
QVG-Pro	5.20×	**31.562**	**0.923**	**0.069**	98.00	74.15	97.96	69.45
QVG	**7.05**×	29.174	0.882	0.094	97.98	73.87	97.90	69.80
HY-WorldPlay-8B		*INT4 KV Cache, 480p*			97.92	74.33	97.90	69.85
RTN	3.55×	33.634	0.948	0.056	97.98	74.26	97.87	70.13
KIVI	3.55×	33.768	0.950	0.055	97.95	74.30	97.95	69.92
Quarot	3.55×	33.997	0.951	0.053	97.97	74.33	97.90	69.76
QVG-Pro	3.15×	**35.109**	**0.960**	**0.048**	97.93	74.30	97.88	69.45
QVG	**3.75**×	34.454	0.954	0.051	97.96	74.23	97.90	69.66

Quantized Keys Steal Attention

历史KV已量化，当前chunk KV保持高精度

二者在同一个softmax中竞争概率质量

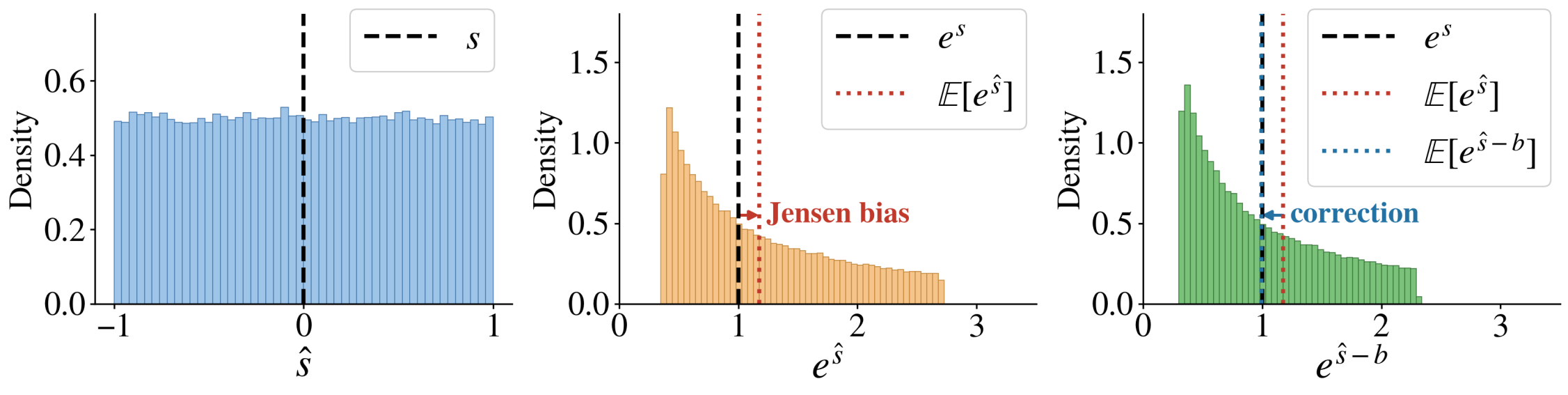
零均值logit噪声经过exp后产生正偏

$$s_i = \frac{q^\top k_i}{\sqrt{d}}, \qquad p_i = \frac{e^{s_i}}{\sum_{j=1}^{N} e^{s_j}}. \tag{2}$$

$$Z_{\mathcal{S}} = \sum_{i \in \mathcal{S}} e^{s_i}, \qquad Z_{\mathcal{R}} = \sum_{i \in \mathcal{R}} e^{s_i}, \qquad Z = Z_{\mathcal{S}} + Z_{\mathcal{R}}. \tag{3}$$

$$P_{\mathcal{S}} = \sum_{i \in \mathcal{S}} p_i = \frac{Z_{\mathcal{S}}}{Z_{\mathcal{S}} + Z_{\mathcal{R}}}, \tag{4}$$

Jensen Bias：指数放大噪声



score噪声δ的均值可以为0

但E[exp(δ)] ≥ 1，历史块partition sum膨胀

partition sum无偏与softmax概率无偏不同

Score Correction的推导与实现

独立均匀舍入噪声：方差由query与量化步长决定

softmax前仅对历史logit减去bi

Sinh计算困难，通过泰勒展开二阶近似

$$\sigma_i^2 = \mathrm{Var}(\delta_i) = \frac{1}{12\,d}\sum_{c=1}^{d} q_c^2\,\Delta_{i,c}^2. \tag{7}$$

$$e^{s_i - b_i}\cdot \mathbb{E}\left[e^{\delta_i}\right] \stackrel{!}{=} e^{s_i} \quad \Longrightarrow \quad \boxed{b_i = \log \mathbb{E}\left[e^{\delta_i}\right].} \tag{10}$$

$$\boxed{b_i = \sum_{c=1}^{d} \log\left(\frac{\sinh\left(\frac{q_c\,\Delta_{i,c}}{2\sqrt{d}}\right)}{\frac{q_c\,\Delta_{i,c}}{2\sqrt{d}}}\right)} \tag{12}$$

$$\boxed{b_i \approx \frac{1}{24\,d}\sum_{c=1}^{d} q_c^2\,\Delta_{i,c}^2} \tag{13}$$

偏差校正：可与QVG组合

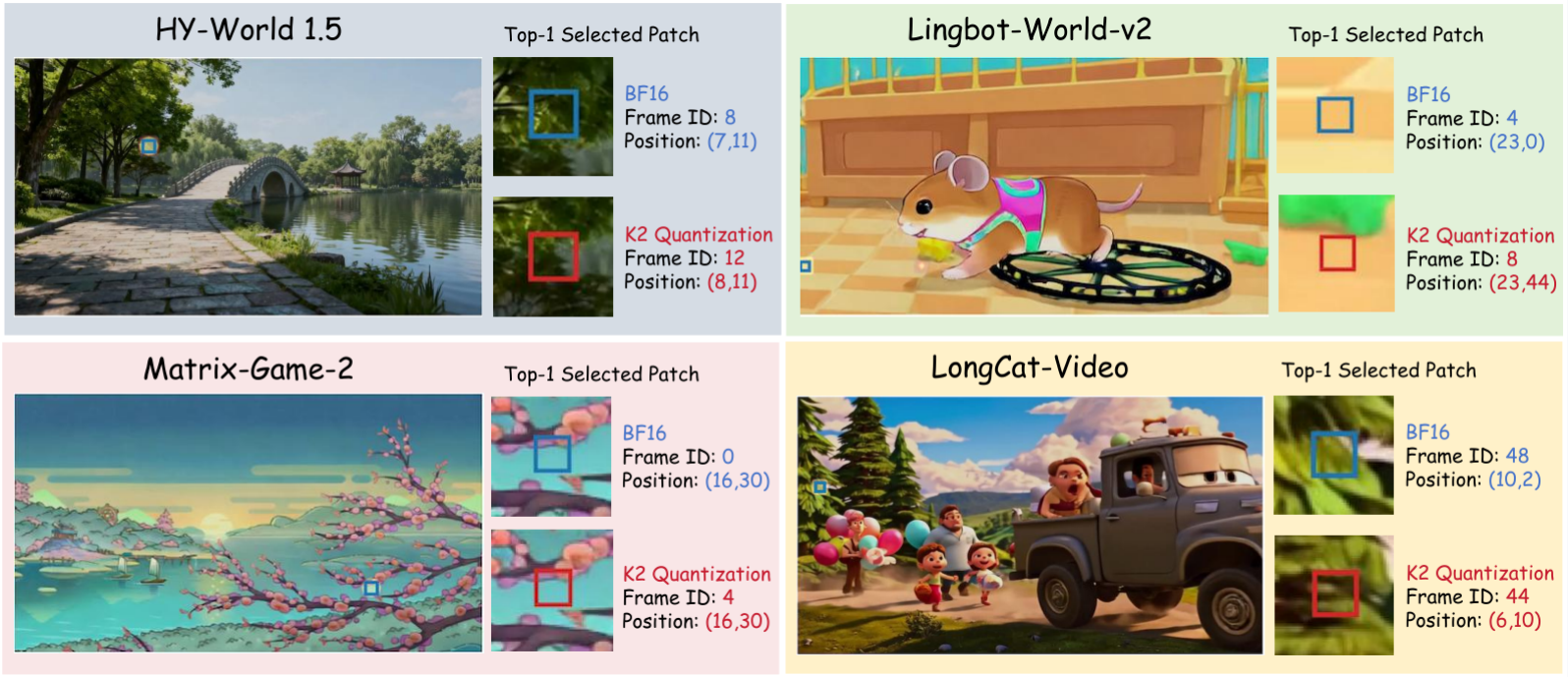
校正不增加缓存metadata，
但增加score计算

论文报告约5%端到端开销，
相对量化基线

Table 1: Effect of the proposed correction for MAGI-1, SkyReels-V2, and HY-WorldPlay. The correction consistently improves fidelity (PSNR, SSIM, LPIPS) and perceptual quality (VBench), recovering much of the degradation introduced by quantization. RTN and QuaRot+RTN rows use an effective bitwidth of 2.75 at INT2; QVG rows on MAGI-1 use the default QVG configuration, which yields an effective bitwidth of approximately 2.52. Standard errors for all metrics are reported in Tables 2 and 5.

Model	Quantization scheme	Precision	With correction	PSNR↑	SSIM↑	LPIPS↓	VBench↑
MAGI-1	—	BF16		–	–	–	78.27
	RTN	INT2	×	23.17	0.799	0.195	76.67
			✓	24.08	0.828	0.131	77.86
	QuaRot+RTN	INT2	×	17.10	0.630	0.453	70.24
			✓	22.97	0.801	0.165	78.02
	QVG	INT2	×	23.01	0.826	0.132	77.81
			✓	**25.29**	**0.856**	**0.107**	**78.23**
SkyReels-V2	—	BF16		–	–	–	78.89
	RTN	INT2	×	18.38	0.670	0.383	68.83
			✓	18.39	0.743	0.269	**80.04**
	QuaRot+RTN	INT2	×	19.20	0.708	0.319	71.44
			✓	**20.42**	**0.784**	**0.202**	78.58
HY-WorldPlay	RTN	INT2	×	16.96	0.573	0.390	–
			✓	17.15	0.577	0.311	–
	QuaRot+RTN	INT2	×	17.16	0.575	0.376	–
			✓	**18.27**	**0.616**	**0.273**	–

QuantWM：Key误差与时空选择



小Key MSE仍可能改变Attention排名

Query对不同通道的误差敏感性不同

目标转向logit扰动与时序一致性

为什么优化QK logits，而不只优化K的MSE？

Key MSE：各方向一视同仁

$$L_K \propto \|\|\|E\|\|\|^2, E = \hat{K} - K$$

Logit误差：Query决定敏感方向，$M = E_q[qq^T]$

Δs = qᵀe / √d ； $E_q[(\Delta s)^2] = e^T M e / d$

同样的MSE，Attention影响可以相差很大

q = (10, 0)：e_1 = (ε, 0)影响分数，e_2 = (0, ε)不影响

QSAC：按量化后的logit风险选中心

历史Query二阶统计估计通道敏感度

Query加权距离先选top-M候选中心

用组残差范围与敏感度重新评分

$$\mathbf{M}_h^{t-1} = \frac{1}{N_{<t}} \sum_{\tau<t} \sum_i \mathbf{q}_{\tau,i,h} \mathbf{q}_{\tau,i,h}^\top. \tag{6}$$

$$\mathbb{E}_{\mathbf{q}} \left[(\mathbf{q}^\top \mathbf{e})^2 \right] = \mathbf{e}^\top \mathbf{M}_h^{t-1} \mathbf{e}, \tag{7}$$

$$w_{h,c} = \frac{[\mathbf{M}_h^{t-1}]_{cc}}{\frac{1}{d} \sum_{c'} [\mathbf{M}_h^{t-1}]_{c'c'}}. \tag{9}$$

$$S(\mathbf{k}_i, \boldsymbol{\mu}_j) = \sum_g \left(\sum_{c \in \mathcal{G}_g} w_{h,c} \right) R_g(\mathbf{r}_{ij})^2, \qquad z_i = \arg\min_{j \in \mathcal{C}_i} S(\mathbf{k}_i, \boldsymbol{\mu}_j). \tag{14}$$

w：Query通道敏感度　R：组内残差范围
较小的加权范围平方，意味着更低的量化logit风险

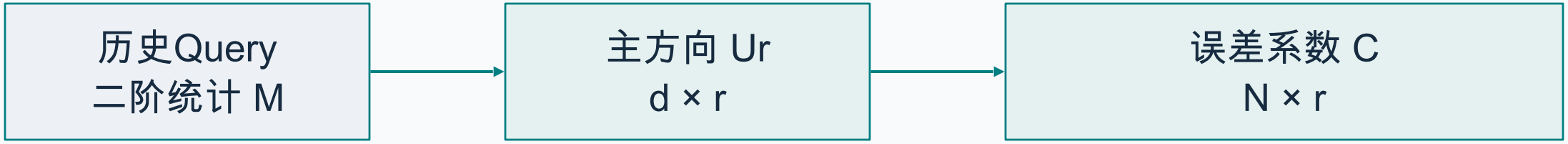
PSAC：保留主Query子空间中的误差

历史Query决定重要方向
投影误差，仅保留r维系数
在Key与logit中补回该分量

$$\mathbf{M}_h^{t-1} = \mathbf{U}_h \mathbf{\Lambda}_h \mathbf{U}_h^\top, \qquad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d. \tag{17}$$

$$\mathbf{C} = \mathbf{E}\mathbf{U}_r, \tag{18}$$

$$\hat{\mathbf{K}} = \hat{\mathbf{K}}_0 + \mathbf{C}\mathbf{U}_r^\top, \qquad \mathbf{Q}\hat{\mathbf{K}}^\top = \mathbf{Q}\hat{\mathbf{K}}_0^\top + \mathbf{Q}\mathbf{U}_r\mathbf{C}^\top. \tag{19}$$



低秩表示节省存储

Results

Table 3: Visual quality and VBench comparison results of **Quant*WM*** and baselines on 480p videos. **Quant*WM*** significantly improves visual quality while maintaining strong VBench performance.

Model	Method	Frame-level Metrics			*VBench*				
		PSNR ↑	**SSIM ↑**	**LPIPS ↓**	*Subject ↑*	*Background ↑*	*Aesthetic ↑*	*Image ↑*	***Avg.* ↑**
Matrix-Game-2	BF16	∞	1.0000	0.0000	0.9104	0.9423	0.5846	0.6952	0.7831
	KIVI	19.298	0.6557	0.1797	0.9097	0.9418	0.5841	0.6914	0.7817
	QVG	17.749	0.5651	0.2851	0.8894	0.9320	0.5831	0.6603	0.7662
	Ours	**20.615**	**0.7212**	**0.1443**	**0.9109**	**0.9421**	**0.5883**	**0.6933**	**0.7836**
LingBot-v2	BF16	∞	1.0000	0.0000	0.8877	0.9126	0.5949	0.6962	0.7728
	KIVI	14.799	0.5950	0.2508	0.8869	**0.9137**	0.5883	0.6892	0.7695
	QVG	14.414	0.5726	0.2779	0.8860	0.9127	0.5912	0.6929	0.7707
	Ours	**16.138**	**0.6688**	**0.1893**	**0.8894**	0.9134	**0.5947**	**0.6953**	**0.7732**
HY-World 1.5	BF16	∞	1.0000	0.0000	0.9588	0.9520	0.6378	0.7209	0.8174
	KIVI	17.677	0.7165	0.1540	0.9520	0.9494	0.6362	0.7049	0.8106
	QVG	16.118	0.6559	0.2114	0.9503	0.9488	**0.6394**	0.7122	0.8127
	Ours	**18.449**	**0.7661**	**0.1082**	**0.9600**	**0.9528**	0.6366	**0.7196**	**0.8173**
LongCat-Video	BF16	∞	1.0000	0.0000	0.9747	0.9605	0.6148	0.6993	0.8123
	KIVI	24.448	0.9180	0.0399	**0.9754**	0.9610	0.6141	0.6695	0.8050
	QVG	20.683	0.8556	0.0784	0.9739	0.9603	0.6144	0.7001	0.8122
	Ours	**25.508**	**0.9286**	**0.0332**	0.9748	**0.9615**	**0.6154**	**0.7002**	**0.8130**
Causal-Forcing	BF16	∞	1.0000	0.0000	0.9327	0.9421	0.6348	0.7028	0.8031
	KIVI	15.585	0.7201	0.1775	0.9334	0.9426	0.6343	0.7034	0.8034
	QVG	14.887	0.6773	0.2185	0.9311	0.9412	0.6317	0.6993	0.8008
	Ours	**16.283**	**0.7464**	**0.1554**	**0.9342**	**0.9427**	**0.6354**	**0.7044**	**0.8042**

三篇论文：优化目标与额外成本

方法	主要目标	额外成本
QVG	减小残差量化误差	中心、索引、聚类与重建
Bias Correction	校正指数权重偏差	query组范数与score校正
QuantWM	保护logit与时空选择	历史统计、主子空间与系数